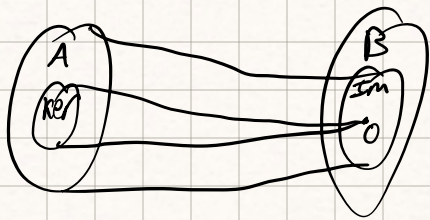


Last time: rank-nullity thm

$$L: A \rightarrow B \quad \dim(A) = \underbrace{\dim(\ker L)}_{\text{nullity}} + \underbrace{\dim(\text{Im } L)}_{\text{rank}}$$



EX: $L: V \rightarrow V$ is linear assume $\dim(\ker L)$ is finite and $\dim(\text{Im } L)$ is finite

What can we say abt the dim of V ?

$$\dim(V) = \dim(\ker L) + \dim(\text{Im } L) \quad \begin{array}{l} \text{RHS is finite} \\ \text{so LHS must also be finite} \end{array}$$

EX:
$$\begin{array}{l} 3x + 2y = 4 \\ 2y + z = 2 \\ x + y + z = 1 \end{array} \rightarrow \begin{bmatrix} 3 & 2 & 0 \\ 0 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix}$$

add row 1 to row 2

$$\begin{bmatrix} 3 & 2 & 0 \\ 3 & 4 & 1 \\ 1 & 1 & 1 \end{bmatrix} \text{ corresponds to } (3x + 2y) + (2y + z) \text{ and } 4 + 2$$

$$\begin{array}{l} 3x + 2y = 4 \\ 3x + 4y + z = 6 \\ x + y + z = 1 \end{array} \quad \begin{bmatrix} 3 & 2 & 0 \\ 3 & 4 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \\ 1 \end{bmatrix}$$

EX:
$$\left[\begin{array}{ccc|c} 3 & 2 & 0 & 4 \\ 0 & 2 & 1 & 2 \\ 1 & 1 & 1 & 1 \end{array} \right] \rightarrow \dots \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & ? \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & ? \end{array} \right] \quad \begin{array}{l} x = ? \\ y = 0 \\ z = ? \end{array}$$

Consequences: system of equations has a unique solution iff L is invertible

if L has a nontrivial kernel then the system has more than 1 solution

Ex: $2x + 4y = 6$
 $x + 2y = 3$ $\underbrace{\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}}_A \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \end{bmatrix}$ what is the kernel?

$$\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0 \quad 2x + 4y = 0 \quad x = -2y \quad \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$\dim(\ker A) = 1 \iff 1$ free variable in solutions of the system

in general: if $L: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is linear and $\dim(\ker L) = k$ then the system $Lx = b$ has either no solutions ($b \notin \text{Im } L$) or k linearly independent solutions

Proof: if $Lx = b$ and $y \in \ker L$, then $Ly = 0 \Rightarrow Lx + Ly = b + 0$
then $L(x+y) = b$

Thm: Let $L: A \rightarrow B$ be linear the following are equivalent

- 1) L is invertible
- 2) L is surjective and injective
- 3) $\det(L) \neq 0$
- 4) L takes a basis to a basis
- 5) L is square and its columns form a basis for B
- 6) L row reduces to I
- 7) L has a trivial kernel $\dim(A) = \dim(B)$
- 8) $Lx = b$ has exactly 1 sol for all $b \in B$

Recall if $L, M: A \rightarrow B$ then $L+M: A \rightarrow B$

$$(L+M)(a) := L(a) + M(a)$$

in coords this is just adding matrices

also kL is a linear map $(kL)(a) = k(L(a))$ in coordinates this is "multiply all entries by k "

$$3 \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 9 & 12 \end{bmatrix}$$

Linear maps $A \rightarrow B$ have

- addition
- scalar mult (scalars from B)
- $0: A \rightarrow B$ $0(a) := 0$
 $\underbrace{\quad}_{\text{zero linear map}} \quad \underbrace{\quad}_{0 \text{ vector in } B}$

Linear functions $A \rightarrow B$ form a vector space